

Project Proposal

Real Time Drawing Recognizer Using Google Quick, Draw! Dataset

1. Dataset Location

The project will use the Google Quick, Draw! Dataset, publicly available [here](#).

2. Dataset Description

The Quick, Draw! Dataset was originally collected by Google as part of an online game where players were asked to draw objects within 20 seconds. The full dataset contains over 50 million drawings across 345 categories (e.g., cat, house, bicycle, sun). Each drawing is stored as a sequence of pen strokes and is available as 28x28 pixel bitmap in .npy format. This makes it straightforward to feed into a neural network.

3. Planned changes to the Dataset

Two modifications will be made to the original dataset.

- Category Reduction: Only 15-20 categories will be selected. This reduces the total data to ~1-2GB and keeps training time reasonable for a standard laptop.
- Sample Capping: A maximum of 10,000 samples per category will be used for training and 2,000 samples for validation. This ensures category balance and prevents any single category from dominating training. The dataset contains far more samples per category than needed for good accuracy.

4. Benchmark Results

Google has published benchmark results for this dataset. Their own classifier achieves ~92% top 1 accuracy across all 345 categories. Since the project uses only 15-20 categories, a target accuracy of 90-95% is realistic and will serve as the performance goal.

5. Neural Network Structure

The project will use a Convolutional Neural Network. The planned architecture is:

- Input layer: 28x28x1 image
- Conv2D (32 filters, 3x3, ReLU)
- Conv2D (64 filters, 3x3, ReLU)
- Conv2D (128 filters, 3x3, ReLU)
- Flatten -> Dense (256, ReLU) -> Dropout (0.5)
- Output Dense

6. Novel Neural Network Structures

Novel Neural Network is planned as the primary model. However, as a stretch goal, a Long Short-Term Memory (LSTM) network may be explored as an alternative, since Quick, Draw! Drawings are also available as ordered stroke sequences. An LSTM could treat a drawing as a time series of pen movements rather than a static image, which may capture temporal drawing patterns that a Convolutional Neural Network can miss.

7. Pretrained Neural Networks in Keras

The project will not rely on a pretrained Keras model as the primary approach, since the drawings are 28x28 grayscale images, much smaller and simpler than the 224x224 RGB images that models like VGG16 were trained on. However, MobileNetV2 with input resizing may be tested as a comparison experiment to measure the effect of transfer learning on this type of data.

8. Expected Outcomes

The project has three expected deliverables:

- Trained Convolutional Neural Network: A CNN trained to classify 15-20 Quick, Draw! Categories with $\geq 90\%$ validation accuracy.
- Interactive Demo: A simple browser-based GUI where a user draws something with their mouse and the model returns its top 3 predictions in real time. This demo will be shown during presentation.
- Analysis Report: A summary of training/validation accuracy curves, a confusion matrix across categories, and comparison against the Google benchmark. Failure cases (drawing the model misclassifies) will be visualized and discussed.